Lab 3: Component Modelling & Architectural Pattern Selection – Written Justification

UE24CS341A Software Engineering Lab | Team 001 | N Lakshanyaa (PES1UG24AM169) | Smart Lab Equipment Slot Reservation Portal – Epic 1: Check Slot Availability (UC-05)

Architecture Selection: We chose Layered Architecture for the Smart Lab Equipment Slot Reservation Portal.

Architectural Choice

Layered Architecture, with a Presentation Layer (Student Portal UI), a Business Layer (Reservation Manager, Availability Checker, Calibration Service, Token Generator) and a Data Layer (Database). Dependencies run only downward, through the interfaces IReservationService, IAvailabilityCheck, ICalibrationCheck, ITokenService and three data-access interfaces. The Reservation Manager plays the role of the Order Manager in the handout, and the Calibration Service is its main partner component.

Style	Pros / Cons for this portal	Verdict
Layered	Clear separation of concerns; one authoritative data layer; some call overhead.	Best fit
Microservices	Independent scaling, but network latency and split data make double-booking hard to prevent.	Over-engineered
Client-Server	Simple and centralised, but no internal structure; single point of failure.	Too coarse

Two Reasons

- **The Epic 1 stories form a fixed pipeline.** A request goes from selecting equipment and a slot (1.1), to the calibration check (1.2), overlap detection (1.3), the 2-hour and 7-day rules (1.4), slot locking with a token (1.5) and the confirmation (1.6). This is a top-down flow that maps directly onto UI, business logic and data, and each component has one clear responsibility that a team member can build and test independently.
- **Double-booking must be prevented from one source of truth.** FR-003 requires that two overlapping requests for the same equipment never both succeed. A single Data Layer that all business components read through lets the overlap check and the slot lock work on the same reservation records. Microservices with separate databases would introduce data-consistency problems, and the portal's campus-scale load does not justify the extra operational complexity.

Security Advantage

Students never reach the database directly. Every request passes through the Reservation Manager, which validates input, applies the calibration, 2-hour and 7-day rules and generates the reservation token before anything is written. Only the Data Layer holds database credentials and runs parameterised queries, so access is controlled at one place and a compromised UI cannot bypass the business rules.

Performance Benefit

The components in the Business and Data Layers run in the same deployment, so the availability check (FR-001) is an in-process call followed by one indexed query on equipment ID and time range. There are no network hops between services, which keeps the real-time availability view fast. Frequently requested slot data can also be cached in the Business Layer without changing the UI or database.